

Background

ABC Institute currently conducts examinations manually using paper-based methods. The process is time-consuming, prone to errors, and difficult to manage for large numbers of students. The institute now plans to develop a **web-based Online Examination Management System** using ASP.NET and SQL Server.

The system should allow:

- **Admin** to create exams, add questions, and manage users
- **Students** to log in, take exams, and view results
- **System** to automatically evaluate objective-type questions and generate reports

Based on the above scenario, answer the following:

1. Propose a suitable architecture for this web application and explain each layer
2. Which ASP.NET State Management technique would you use to maintain a student's session during the exam? Justify your answer
3. The Admin needs to create a new exam with fields like "Exam Name" and "Total Marks". Which ASP.NET Validation Controls will you use to ensure valid input?
4. Write a C# ADO.NET code snippet to insert a new exam record into the SQL Server database
5. Explain how the system can automatically evaluate objective-type questions and generate results